Prime-IC relation-solving tournament v2: preflight report

Qualification start: 2026-11-01 04:00 UTC
Frozen protocol: `IC-LINE-LAYOUT-E2E-FP20-V2-20261101`
Tested source: `ea8e9ef71c3c82027dd9e01535302725519c3fca`
Outcome: the single L2 qualification failed; the scientific tournament was not dispatched.

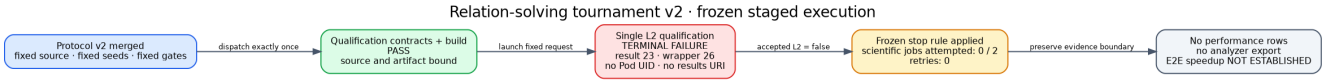


Figure 1: Frozen staged execution and terminal evidence

The diagram is an execution-accounting graphic, not a performance graph. Its editable source is `failure-flow.dot`, with an SVG copy at `failure-flow.svg`.

Requested scope and evidence status

Requested or required item	Status	Evidence
Merge a frozen v2 protocol before data	Complete	PR #3151, merge `ea8e9ef71c3c82027dd9e01535302725519c3fca`
Run one fixed L2 qualification	Terminal failure	Workflow 41271383665
Launch two scientific sessions only after accepted L2	Not attempted	Frozen fail-closed rule applied
Analyze complete relation-solving pipelines	Unavailable	No results URI or authenticated session directory
Preserve zero outcome-driven retries	Complete	One qualification attempt; zero retries, replacement seeds or scientific jobs
Establish an end-to-end ECDLP speedup	Not established	Zero admitted performance rows or comparisons

The execution counts are qualification planned/attempted/terminal/accepted-L2 `1/1/1/0` and scientific jobs conditionally planned/attempted `2/0`. This is a terminal observation of the frozen v2 design, not candidate data.

Identity-bound terminal receipt

The workflow contract and build jobs passed. Artifact `13069693394` (`isolab-build-41271383665-1`) contains the build bound to the tested source; its 82,666,555-byte archive digest is `sha256:29ab639216eadbc2041bd92467d98fb7a7d774f2bd0e0990b0f9f095a3bddd4d`. The experiment job downloaded that exact digest, published immutable image `sha256:3c4ad8bf8ee1bd1fdc327cb4c0e2597a1fc5ef1cdd889aa21e1b1294bdd22914`, and launched the fixed request with native binary SHA-256 `73362eed4165d6db20fe1d398baa8fabefa2497aff6e78dab9ed8f31af31dd01`.

Receipt field	Observed value
Workflow source revision	`e443a60f00839aa84a7734053b8d6b5b0bad0f40`
Tested source revision	`ea8e9ef71c3c82027dd9e01535302725519c3fca`
Qualification spec	`taskq/deploy/eks-karpenter/l2-validation.json`
Job / UID	`gh-41271383665-1` / `06bc6798-a65b-4805-9e43-47f16be8d97f`
Launcher ID	`c4090885-9236-4b36-8dba-8cea6f746655`
Request SHA-256	`9e4c461dd35f5301360efb1a67351d609773 5857d79cc79ae055427684f70971`
Terminal state	`failed`; remote-result exit 23; wrapper exit 26
Successful Pod UID / results URI	absent / absent

The receipt proves a terminal infrastructure-qualification failure. It does not identify a payload exit code, an L2 predicate or a scientific result. The older v1 diagnosis found exit 42 at `l2_isolation_gate` for two different fixed Jobs; that classification is not transferable to this v2 qualification.

Stopping rule and missing export

Protocol SHA-256 `0dc2be59cd4860fae6775d0dff823c1025d4fa999b2c9550bd4d7e19c030f92b` required a manifest-verified accepted-L2 qualification before scientific dispatch. The failed launcher receipt has no successful Pod UID or results URI, so it cannot satisfy that gate. Both scientific sessions were therefore left unsubmitted. There was no GitHub rerun, replacement seed, host selection, factor-base retuning or threshold change.

The frozen analyzer accepts two authenticated `ecbench.session/v1` directories. This run produced none. Consequently there is no honest `experiment.export/v1`, replay receipt, A/A comparison, positive-control comparison, candidate interval or numeric graph to publish. A flow diagram is used because fewer than two comparable values—and in fact no timing values—are available. The scoreboard, leaderboard and `benchmarks/` archive are unchanged.

Evidence limits and next discriminating step

The public launcher receipt is the highest evidence available here. It carries the fixed request, build and Job identities but no payload classification or manifest-verified L2 detail. It is sufficient to apply the frozen stop rule and insufficient to attribute the exact infrastructure cause. No candidate repair or algorithmic conclusion follows from it.

The existing problem record requires either narrowly authorized, manifest-bound diagnosis or a separately reviewed infrastructure qualification. That future work must not rewrite this consumed qualification or broaden its evidence into a speed claim. A new scientific evidence version would still need the unchanged L2, replay, exact-operation, A/A, positive-control and two-session confidence gates.

Sources checked

- `PROTOCOL-V2.md` and `protocol-v2.json`: frozen staged rule, claims and analysis boundary.
- `taskq/deploy/eks-karpenter/HARNESS.md`: request identity, launcher result and private-payload boundary.
- `taskq/deploy/eks-karpenter/ACCESS.md`: short-lived workflow identity and scoped artifact route.
- GitHub workflow 41271383665: exact build, publication and terminal public receipt.
- `V2-PREFLIGHT-OBSERVATION-20261101.md`: concise human observation with the machine record beside it.

Claim boundary

Relation-yield pruning instrumentation from PR #3091 is separately merged engineering work. This preflight contains no scientific relation-solving session and no admitted end-to-end timing. The full ECDLP index-calculus speedup claim remains unestablished.